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Research paper

Intra-articular delivery of genipin cross-linked chitosan microspheres of flurbiprofen: Preparation, characterization, *in vitro* and *in vivo* studies

Jitendra Kawadkar, Meenakshi Kanwar Chauhan *

Department of Pharmaceutics, University of Delhi, New Delhi, India

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ABSTRACT

In this study, genipin cross-linked chitosan microspheres of flurbiprofen for intra-articular (i.a.) delivery were prepared. Emulsion-cross-linking method was used to prepare the microspheres using different concentrations of genipin and drug-to-polymer ratios. The mean particle size was found to be in the range of 5.18–9.74 μm with good % drug entrapment up to 80.97%. SEM indicated the spherical shape with smooth surface of drug-loaded cross-linked microspheres. FTIR confirmed cross-linking of genipin with chitosan and the absence of chemical interactions between drug, polymer, and cross-linker, which was further confirmed by TGA showing unaltered melting point of entrapped drug. DSC and XRD revealed the molecular dispersion of drug within microspheres. The optimized microspheres were able to release the drug for more than 108 h. The biocompatibility of the microspheres in the rat (Sprague-Dawley) knee joints was confirmed by histopathology. The results of pharmacokinetic study, decreased K_a and T_{max}, lower C_{max} and $AUC_{(0-24)}$ and delayed MRT, suggested the significant extended release of flurbiprofen from microspheres in comparison with its solution at $P < 0.05$. The recovery of flurbiprofen as the percent of administered dose followed by 24 h after i.a. injection of microspheres was found to be 8.7 folds higher than its solution.

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1. Introduction

Recently, considerable interest has been paid to employ the intra-articular (i.a.) drug delivery in the treatment of arthritic disorders such as rheumatoid arthritis, degenerative joint disease, osteoarthritis, and ankylosing spondylitis. It sufficiently helps the drug to perform local action within the joint cavity by attaining the maximum concentration and minimum exposure to inappropriate sites [1]. The drug delivery from oral and parenteral routs provides low therapeutic concentration in the joint cavity. The intra-articularly injected drug solutions are rapidly cleared from the joint spaces and needed frequent injections to maintain the drug concentration within the joints. Therefore, polymer-based microparticulate drug delivery systems (microspheres/microparticles), for the purpose of controlled drug delivery for desired period of time, have been extensively studied [2–9]. In order to achieve controlled drug release, chitosan polymer, a deacetylated derivative of chitin, has been commonly used in the formulation of microparticulate drug delivery [10,11]. Biocompatibility and biodegradability of chitosan have made it appropriate biopolymer for biomedical application. [12]. The cross-link increases the potential of chitosan for drug release over an extended period of time [13]. Therefore, considering biocompatibility, many investigations have been undergone to develop new reagents to cross-link chitosan. The chemically synthesized cross-linkers most commonly used are glutaraldehyde, formaldehyde, tripolyphosphate, ethylene glycol, diglycidyl ether and disulfide. However, these compounds have physiological toxicity [14,15]. Currently, genipin, a natural cross-linker, is obtained from geniposide which is isolated from the fruits of *Gardenia jasminoides* Ellis. It is a white crystalline, water soluble powder and bifunctional cross-linker [16]. It reacts with amine groups of biopolymers and changes them in dark blue color [17]. This naturally occurring cross-linker is approximately 10,000 times less toxic as compared to glutaraldehyde [18]. Mi et al. [19] also reported, using injectable chitosan-based implant in rodent model, that chitosan microspheres cross-linked with genipin are histologically better biocompatible and show slower degradation rate than cross-linked with glutaraldehyde. For i.a. drug delivery, the glutaraldehyde cross-linked microspheres of polysaccharides such as albumin, chitosan, and gelatin have been prepared but the use of this cross-linker is always doubtful in case of biocompatibility.

Flurbiprofen, a phenylalkanoic acid derivative, is one of the common drugs that are frequently used in the treatment of arthritic disorders. It is a non-steroidal anti-inflammatory, anti-pyretic, and analgesic agent that inhibits the enzymatic activity of cyclooxygenase, leading to the suppression of prostaglandin

* Corresponding author. NDDS Research Lab, Department of Pharmaceutics, Delhi Institute of Pharmaceutical Sciences and Research, University of Delhi, Sector-3, Pushp Vihar, M.B. Road, New Delhi 110 017, India. Tel.: +91 8750545349; fax: +91 1129554503.

E-mail address: meenakshindds@gmail.com (M.K. Chauhan).

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synthesis [20]. It is given orally in multiple doses (50–75 mg single dose for 3–4 times daily), because of its short elimination half-life (4 h), in order to achieve and maintain therapeutic concentration [21]. In addition, the oral administration of this drug causes toxicity at gastrointestinal level [22]. Because, the orally administered drugs are not able to achieve therapeutic concentration in joint cavity, therefore, a controlled release formulation of this drug for i.a. delivery is highly desired.

The aforementioned reports encouraged us to prepare microspheres using this herbal reagent to cross-link the chitosan for i.a. administration. In this study, we present the characterization of genipin cross-linked chitosan microspheres of flurbiprofen. The effect of genipin concentrations on physicochemical properties of microspheres was investigated. Biocompatibility of these microspheres was examined in the rat knee joints. The pharmacokinetic parameters of flurbiprofen solution and its microspheres after i.a. administration in the arthritic rat knee joints were also evaluated.

2. Materials and methods

2.1. Materials

Chitosan (MW $\sim 3.425 \times 10^5$, degree of deacetylation $\sim 85\%$) and carrageenan were purchased from Sigma–Aldrich Co., USA. Flurbiprofen was obtained from Sun Pharmaceutical Industries Ltd., India. Genipin was purchased from Challenge Bioproducts Co. Ltd., Taiwan. All chemicals and reagents used were of analytical grade, and solvents were of high-performance liquid chromatography (HPLC) grade.

2.2. Preparation of chitosan microspheres

Chitosan microspheres were prepared by an emulsification-cross-linking method using genipin as a cross-linker. Chitosan (90 mg) was dissolved in 3 mL of 0.5 M aqueous acetic acid solution, containing Tween-20 (2.5% w/v). Flurbiprofen (30 mg) was mixed with chitosan solution using a bench top homogenizer (Plytron® PT-MR3100D, Kinametica, Switzerland) at 10,000 rpm for 2 min. The aqueous phase was dispersed into a mixture of liquid paraffin (light) (40 mL) and Span-80 (2% w/v), and homogenization was continued at 8000 rpm for 10 min. The emulsion formed was heated at 37 °C with continuous stirring at 800 rpm using a magnetic stirrer (5MLH, REMI, India). Genipin solution (2 mL of 70% alcohol) was added dropwise into the emulsion to cross-link the microspheres. The cross-linking reaction was allowed for 2 h. The microspheres were then separated out, washed three times with petroleum ether and acetone followed by isopropyl alcohol by centrifugation process. The prepared microspheres were dried under vacuum at room temperature. The representative formulations are given in Table 1.

2.3. Characterization of chitosan microspheres

2.3.1. Drug entrapment

The microspheres (20 mg) were digested in HCl (3 N) at 60 °C for 1 h with stirring using a magnetic stirrer. After cooling up to room temperature, it was then vigorously shaken with methylene chloride to extract flurbiprofen. The organic extract was separated out, evaporated to dryness, dissolved in methanol, and filtered through 0.45 μm pore size Millipore® filters. The amount of flurbiprofen in this clear solution was analyzed UV-spectrophotometrically (Shimadzu 1601, Japan) at 247 nm. Each experiment was performed three times. The drug entrapment was calculated by the following equation:

$$(\%)\ \text{drug entrapment} = \frac{\text{calculated drug content}}{\text{theoretical drug content}} \times 100$$

Table 1
Formulation codes of microspheres.

Genipin (wt.% of polymer)	Drug:polymer ratio		
	1:1	1:2	1:3
0.5	CM-1	CM-4	CM-7
1	CM-2	CM-5	CM-8
2	CM-3	CM-6	CM-9

2.3.2. Particle size distribution

The particle size and size distribution of microspheres were determined by laser diffraction particle size analyzer (Coulter Counter, LS 200, US) using isopropyl alcohol as dispersant. Suspension of microspheres in isopropyl alcohol was prepared. In the sample adopter, stirring at 50% rpm and sonication at 50 mV for 30 s were applied on the suspension. The study was carried out at 8–10% laser obscuration. Size distribution was plotted using the software supplied by the manufacturer. SPAN index (broadness of particle size distribution) is calculated by the following equation:

$$\text{SPAN index} = (D_{90} - D_{10})/(D_{50})$$

In this equation, D_{90}, D_{50}, and D_{10} represent the particle diameters determined at the 90th, 50th, and 10th percentile of undersized particles, respectively [23]. The small SPAN values are indicative of narrow particle size distribution. Three samples of each formulation batch of microspheres were analyzed.

2.3.3. Surface morphology studies

The morphological studies of microspheres were assessed by both light microscope and scanning electron microscope (SEM). The bulk morphology of microspheres during the formulation process was observed by light microscope (BA 400, Motic, China). The investigation of shape and surface of microspheres was carried out by SEM (Carl Zeiss AG-EVO®40, Germany) (at 20 kV). For SEM study, the microspheres were sprinkled onto an adhesive carbon tape and sputter-coated (Balzer Union SCD040 coater) with gold under vacuum.

2.3.4. Swelling measurements

The swelling behavior and water content at equilibrium of uncross-linked and cross-linked chitosan microspheres were measured by swelling of microspheres (100 mg) in 5 mL of deionized water at room temperature for 18 h in screw cape tube with gentle shaking. At preset time intervals, the swollen microspheres were centrifuged and blotted carefully by filter paper followed by a stream of air, blown to remove surface water and weighed immediately on an electronic balance (ABJ 220-4M, Kern & Sohn, Germany). The swelling ratio was calculated from the following:

$$E_{sw}(\%) = [(W_s - W_d)/(W_d)] \times 100$$

where E_{sw} is the percentage water absorption of the microspheres, W_s and W_d are the weights of the microspheres in the swollen and dry state, respectively.

The equilibrium water content (EWC) of the microspheres was calculated using the following equation:

$$\text{EWC}\ (\%) = [(W_c - W_d)/(W_c)] \times 100$$

where W_c is the weight of the swollen microspheres at equilibrium. Each swelling experiment was repeated three times.

2.3.5. Fourier transform infrared spectroscopy (FTIR)

FTIR spectra of genipin, chitosan, pure drug, placebo, and drug-loaded microspheres were recorded using FTIR spectrophotometer

(Perkin-Elmer-Spectrum RX-I, Lambda, USA). The samples were ground with KBr, and pellets were prepared by means of hydraulic pellet press at a pressure of 600 kg/cm^2. The samples were scanned from 4000 to 400 cm^{-1}.

2.3.6. Thermal analysis

Thermogravimetric (TGA) and differential scanning calorimetry (DSC) studies were performed on genipin, pure drug, chitosan, and drug-loaded microspheres cross-linked with at 0.5%, 1%, and 2% genipin. In TGA and DSC studies, samples were placed in aluminum pans and heated at 10 °C/min by using Perkin-Elmer TGA7/DSC7. TGA and DSC measurements were taken with a heating range of 50–450 °C and 40–400 °C, respectively.

2.3.7. X-ray diffractometry (XRD)

XRD of pure drug, chitosan, and drug-loaded microspheres cross-linked with 0.5%, 1%, and 2% genipin was done by using powder X-ray diffractometer (XD-D$_1$, Shimadzu, Japan). XRD pattern of all the samples was recorded with scan speed 4° min^{-1} in the range of 5–50° of 2θ.

2.4. In vitro release studies

Flurbiprofen release from genipin cross-linked chitosan microspheres was observed using pH 7.4 phosphate buffer saline (PBS) as the release medium at 37 ± 1 °C. The microspheres (equivalent to 2.5 mg flurbiprofen) were suspended in PBS (25 mL) containing Tween-80 (1% w/v) in a 50 mL screw capped glass tubes. The suspension was shaken at 100 rpm using horizontal shaking water bath (Scientec, India). At different time intervals, the tubes were centrifuged at 5000 rpm for 5 min, and 2 mL of the supernatant was filtered through 0.45 μm pore size Millipore® filters. The sediment of microspheres was re-dispersed in the same volume of fresh PBS and added into the tubes to maintain sink conditions. The concentration of flurbiprofen in the filtered samples was analyzed by UV-spectrophotometer at 247 nm. The release study was also performed using 2.5 mg of flurbiprofen powder. The release experiments were performed three times in identical manner.

2.5. In vivo studies

2.5.1. Induction of arthritis

All *in vivo* studies, on male Sprague-Dawley rats (0.3–0.35 kg), were approved by the Institutional Animal Ethical Committee with approval No. IAEC/2010/11. The mono-articular arthritis was induced in rats by the method applied by Hansra et al. [24]. Over 21 days, animals received seven i.a. injections (using 30 gauge needle) of sterile 1% w/v carrageenan (20 μL), prepared in physiological saline (i.e., 0.9% NaCl), in the left tibiofemoral joint under ethrane anesthesia. The contra lateral knee joints were used as the control joints. Mono-articular arthritis was developed at 24th day in the rats, and these rats were used for *in vivo* studies.

2.5.2. Histological study of knee joints

The biocompatibility of genipin cross-linked chitosan microspheres with the synovium and the carrageenan induced arthritis in the knee joints both were investigated by histopathology. For the biocompatibility study, microspheres (8–10 mg), sterilized by gamma radiation (source Co-60, dose 25 kGy), suspended in saline (0.1 mL) was injected in the left knee joint, whereas 0.1 mL saline was injected into the right knee joint as control. The rats were sacrificed and joints were removed three days post-injection. They were fixed in 10% phosphate buffered formalin, embedded in paraffin wax, cut in the sagittal plane and sagittal sections (5 μm) and stained with hematoxylin/eosin. Sections were examined under light microscope for inflammatory changes [7].

2.5.3. Pharmacokinetic studies

The arthritic rats were divided into two groups containing eighteen in each. For the blood sampling, under ethrane anesthesia, the femoral arteries of the arthritic rats were cannulated with PE-50 polyethylene tubing. Equivalent to the dose of 2.5 mg/kg of flurbiprofen, i.a. injection of flurbiprofen solution (in 0.2 mL PBS of pH 7.4) and its microspheres (in 0.2 mL PBS) were injected in the left tibiofemoral joint of rats of each group separately. Blood samples (0.15 mL) were obtained via the femoral artery at 0, 15, 30 and 45 min, 1, 2, 4, 6, 8, 12 and 24 h after the injection. An equivalent volume of normal saline was administered into the animals after each blood withdrawal. The blood samples were centrifuged immediately at 5000 rpm for 5 min. The plasma was harvested and stored at −20 °C prior to drug analysis. At 4 and 12 h following i.a. injection, six rats of each group were sacrificed and the rest of the rats were sacrificed at 24 h following i.a. injection. The arthritic knee joints were removed, dried, weighed, and stored at −20 °C for further study.

2.5.4. Analysis of flurbiprofen in rat plasma and tissue

Flurbiprofen in the rat plasma was extracted by adding 2.5 mL of acetonitrile to the sample (50 μL), vortexed, mixed, and centrifuged at 10,000 rpm for 10 min. The supernatant was separated and mixed with 2 mL of methylene chloride. The organic layer was evaporated, and the residue was reconstituted with mobile phase (0.2 mL). The arthritic knee joints (5 g) were homogenized in required amount of methanol. The homogenates were centrifuged at 10,000 rpm for 1 h. The supernatant were collected and treated in the same manner as that for plasma. Analysis of flurbiprofen was carried out by HPLC method as described by Park et al. [25]. The HPLC system was equipped with Shimadzu LC-10 ATvp solvent delivery unit (Shimadzu Corporation, Japan), Shimadzu SIL-10A auto injector, reversed phase Kromasil C$_{18}$ column (200 × 4.6 mm^2 ID, 5 μm), and a Waters 474 fluorescence detector (excitation wavelength of 250 nm and emission wavelength of 285 nm). The mobile phase consisted of acetonitrile–water–phosphoric acid (600:400:0.5, v/v/v); the flow rate was 1 mL/min. The column temperature was ambient and back pressure was 120 kg/cm^2.

2.5.5. Pharmacokinetic parameters

The pharmacokinetic parameters based on non-compartmental approach were estimated by using a computer program, WinNonlin® (Scientific Consulting, Inc., USA; version 1.5). The peak plasma concentration (C_{max}) and its time (T_{max}) were obtained from plasma concentration–time curve. The area under the curve from zero to infinity ($AUC_{0-\infty}$) and mean residence time (MRT) were calculated using trapezoidal method.

2.6. Statistical analysis

Statistical analysis was carried out by SPSS 17 (SPSS Inc., Chicago, USA), using one-way analysis of variance (ANOVA). Dunnett's and Tukey–Kramers's multiple comparison tests were applied in swelling study at a significance level of $P < 0.05$ (significant) and 0.01 (highly significant). Student's *t*-test ($P < 0.05$ = significant, $P < 0.001$ = highly significant) was used for blood kinetic data of two treatments.

3. Results and discussion

3.1. Preparation and characterization of chitosan microspheres

In this study, cross-linked chitosan microspheres were prepared by emulsification-cross-linking method. Genipin, a herbal and non-toxic fruit extract, was used as a cross-linking agent. Chen et al.

Table 2
Mean (±SD) characteristics of different formulation batches ($n = 3$).

Formulation code	Drug entrapment (%)	Volume mean particle size (μm)	SPAN index[a]	EWC (%)	n Value[b]
Uncross-linked	–	–	–	61.29 ± 2.60	0.721
CM-1	64.78 ± 2.33	7.43 ± 1.20	4.94	64.68 ± 2.09	0.682
CM-2	63.39 ± 2.31	5.18 ± 1.07	4.22	55.32 ± 2.12	0.648
CM-3	68.15 ± 3.02	5.46 ± 0.76	2.78	51.13 ± 2.23	0.604
CM-4	67.53 ± 3.15	8.92 ± 1.30	5.21	63.42 ± 3.11	0.692
CM-5	65.24 ± 2.14	5.77 ± 0.63	3.59	53.65 ± 2.03	0.641
CM-6	73.38 ± 2.26	5.91 ± 1.16	2.47	50.64 ± 2.10	0.611
CM-7	74.74 ± 1.97	9.74 ± 0.91	2.14	51.36 ± 3.43	0.685
CM-8	78.71 ± 1.35	7.06 ± 0.75	1.52	43.05 ± 2.17	0.631
CM-9	80.97 ± 2.18	6.14 ± 0.87	1.03	35.59 ± 2.51	0.608

[a] The standard deviation of SPAN index is not shown if less than 0.5.
[b] Diffusional exponent, the standard deviation is not shown if less than 0.5.

[26] reported that the reaction of chitosan with genipin at 37 °C was found to be more rapid because of the higher level of molecular mobility, when compared with lower temperatures. Therefore, in this study, chitosan microspheres were kept for cross-linking with genipin at 37 °C for 2 h. Microspheres were prepared by using different drug-to-polymer ratio and cross-linked at different concentration of genipin.

The drug entrapment was found to be in the range of 63.39 ± 2.31–80.97 ± 2.18% with varying genipin concentration and drug-to-polymer ratio (Table 2). The results showed that the drug entrapment was found to be proportional to the drug-to-polymer ratio. The proportional increase in drug entrapment with an increase in drug-to-polymer ratio was due to increase in viscosity of the polymer solution that reduced the partitioning of drug in external phase, whereas genipin concentrations had not showed any pronounce effect on drug entrapment.

The volume mean particle size and SPAN index of all the formulations are given in Table 2. The volume mean particle size was found to be from 5.18 ± 1.07 μm to 9.74 ± 0.91 μm. The data revealed that the size of microspheres increased with increasing the concentration of polymer which may be attributed to the increased solution viscosity. These results are in agreement with the results of Denkbas et al. [27] who studied the effect of chitosan/acetic acid solution ratio on chitosan microsphere size. But the concentration of genipin did not show any perceptible effect on particle size. On the other hand, it was observed that the values of SPAN index of microspheres were effectively influenced by different concentration of genipin but not the polymer (Fig. 1). The increasing concentration of genipin showed substantial decrease in SPAN index of microspheres. The microspheres cross-linked with low concentration of genipin had wide size distribution (Fig. 1a). This might be due to the aggregation of microspheres as a result of too poor surface solidification by low concentration of cross-linker during the formulation. However, the size distribution has a D_{50} of 7.13 μm and D_{90} of 9.34 μm at highest concentration of cross-linker (Fig. 1c). It indicates that 2% of genipin gives narrow size distribution of microspheres.

Surface morphology and shape of the microspheres were examined by light microscopy and SEM. The light microscopy of chitosan microspheres during the formulation process revealed that the particles are discrete with spherical geometry (Fig. 2a). SEM images of the placebo uncross-linked microspheres (Fig. 2b) and flurbiprofen-loaded cross-linked microspheres (Fig. 2c) indicate that both type of microspheres are spherical, fairly smooth, and with uniform surface. The appearance and shape of the microspheres were not affected by cross-linking and drug loading as low viscosity liquid paraffin as an external phase kept the microspheres in a resistant-free environment along with high speed homogenization.

The swelling of chitosan (pK_a 6.5) in an aqueous environment is due to inherent electrostatic repulsion between the charged groups present on chitosan. [28,29]. The effect of genipin concentration on the swelling capacity of chitosan microspheres in deionized water (pH 6.2) was studied. All the cross-linked microspheres showed low swelling ratio as compared to uncross-linked microspheres (control) that could be due to the formation of molecular linkages (heterocyclic amino linkage) by genipin. The swelling ratio of all the cross-linked microspheres reached a plateau after an immersion period of 8 h but in case of uncross-linked microspheres, the swelling ratio showed an increase with time after 12 h of immersion period. This imply that cross-linking by genipin limits the water uptake due to changed hydrophilicity of chitosan. Fig. 3 indicates that the swelling capacity of microspheres decreases with increase in concentration of genipin. Formulation CM-7 showed an EWC of 51.36 ± 3.43% followed by 43.05 ± 2.17 and 35.59 ± 2.51 for CM-8 and CM-9, respectively ($P < 0.05$), and these values of cross-linked microspheres were significantly lower than EWC of uncross-linked microspheres (61.29 ± 2.6%) ($P < 0.01$). This may be referred that as the concentration of genipin increases, the density of chitosan polymer network also increases. As a result, the available intra- and intermolecular spaces in the microspheres to hold water molecules get reduced. The EWC of all the microspheres are shown in Table 2.

The mechanism of water transport through the genipin cross-linked and uncross-linked microspheres was studied using the first 60% swelling data in the following equation [30]:

$$\frac{M_t}{M_\infty} = kt^n$$

where M_t/M_∞ is the fractional amount of water uptake at time t, k is a diffusional kinetic constant and n is the diffusional exponent. The values of n for formulation CM-7, CM-8, and CM-9 were 0.685 ± 0.003, 0.631 ± 0.002, and 0.608 ± 0.003, respectively ($P < 0.05$), and 0.721 ± 0.003 for uncross-linked microspheres ($P < 0.05$), indicating non-Fickian diffusion ($n > 0.45$). This revealed that the water transport mechanism is mainly due to the combination of both diffusion and erosion controlled, in genipin cross-linked and uncross-linked microspheres. The values of n for cross-linked microspheres were smaller than uncross-linked microspheres, and this implied that genipin gave rigidity to the chitosan by cross-linking. It was observed that the values of n decreased with an increase in cross-linking of chitosan microspheres. The values of n for all the formulation batch of microspheres are shown in Table 2.

The FTIR spectra of genipin, chitosan, flurbiprofen-loaded microspheres cross-linked with genipin and flurbiprofen were compared in Fig. 4A. Chitosan spectrum showed characteristic absorption bands at 3439 cm^{-1}, 1652 cm^{-1}, and 1547 cm^{-1} which correspond to -OH, amide I (C=O), and amide II (NH_2), respectively. The cross-linking of chitosan with genipin was observed by the band shifting of 1547 cm^{-1} to lower wavenumbers attributed to the nucleophilic substitution by the amino group of chitosan on the olefinic carbon atom at C-3 of genipin followed by opening the dihydropyran ring to form heterocyclic amine. The band shifting of 1652 cm^{-1} to higher wavenumbers revealed about the formation of secondary amide group with chitosan as a result of nucleophilic substitution of the ester group on genipin. The association of these intermediate compounds leads to the formation of cross-linked networks with short chains of cross-linking bridges [31].

Chitosan is a polycationic polymer that is likely to interact with acidic drugs but this interaction depends upon the pH of the chitosan solution [32,33]. The FTIR characteristic peaks of flurbiprofen at 1700 cm^{-1}, 1323 cm^{-1}, and 1215 cm^{-1} are attributed to stretching

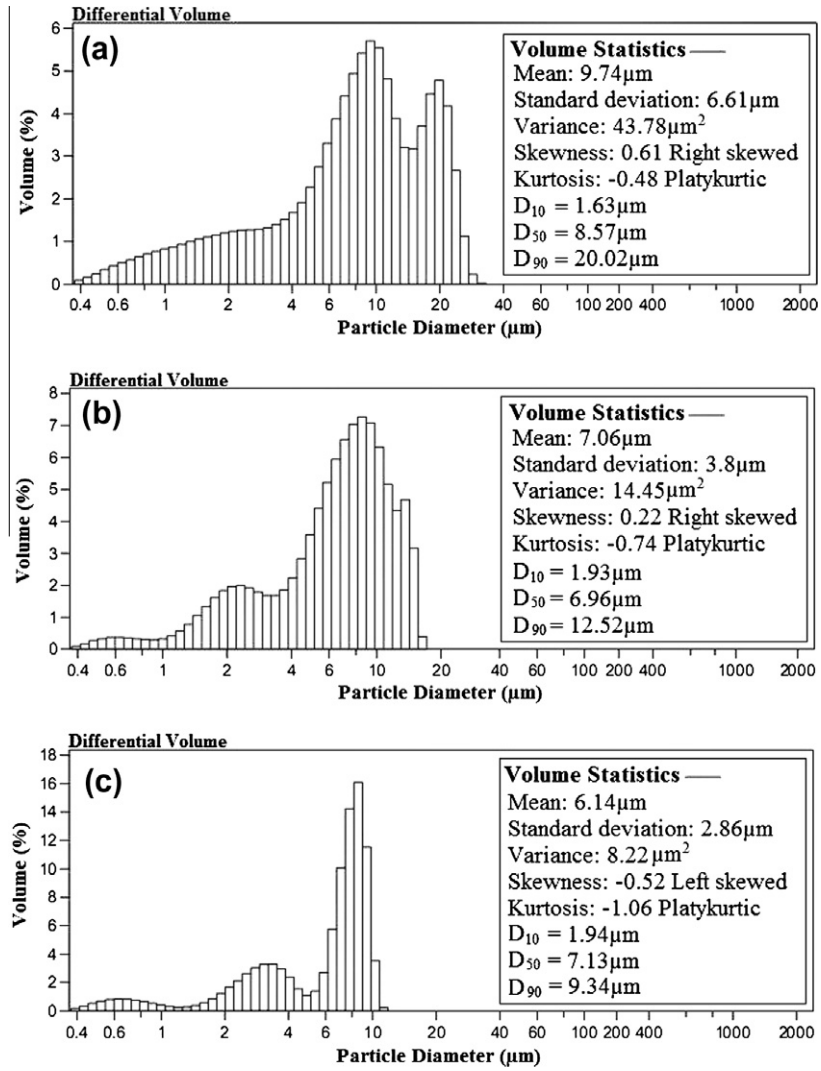


Fig. 1. Volume mean particle size distribution of chitosan microspheres cross-linked with different concentration of genipin: (a) 0.5%, (b) 1%, and (c) 2%.

of —COOH, C—H of methyl and C—F groups, respectively. In the spectrum of flurbiprofen-loaded chitosan microspheres, the characteristics peaks of chitosan along with flurbiprofen were appeared with no significant changes. It might be due to unionized form of drug in acidic solution of chitosan and no reaction between carboxyl group in flurbiprofen and amine group in chitosan.

TGA studies of chitosan, flurbiprofen-loaded chitosan microspheres cross-linked with 0.5%, 1%, and 2% genipin and flurbiprofen are presented in Fig. 4B. The weight loss for chitosan was constant from 120 to 260 °C and beyond this an abrupt weight loss was observed. TGA of chitosan and flurbiprofen showed narrow degradation temperatures, with a maximum temperature at 325 and 280 °C, respectively. Whereas the microspheres have broad degradation pattern with a maximum degradation temperature from 295 to 316 °C which might be due to the decomposition of entrapped flurbiprofen. The sudden decomposition of entrapped flurbiprofen after 200 °C resulted in higher weight loss in microspheres than pure chitosan. The curves of TGA of free flurbiprofen and flurbiprofen-loaded microspheres revealed that the melting point of flurbiprofen (114 °C) was not affected by encapsulation. It assures that no drug interaction with polymer and cross-linker, which is also supported by FTIR studies.

The DSC thermograms of genipin, flurbiprofen, chitosan, and flurbiprofen-loaded chitosan microspheres cross-linked with 0.5%, 1%, and 2% genipin are shown in Fig. 5A. The melting points of genipin and flurbiprofen were revealed by endothermic peaks at 124 °C and 114 °C, respectively. Chitosan microspheres cross-linked with 0.5% genipin showed denaturation enthalpy (ΔH_d) of 36.69 J/g followed by 29.53 J/g and 17.76 J/g for those cross-linked with 1% and 2% genipin, respectively. The ΔH_d of all cross-linked microspheres was found to be lower than the ΔH_d of chitosan (41.59 J/g). This is in accordance with the literature that chitosan with free unsubstituted amine groups showed higher ΔH_d [34]. Thus, the results indicate that ΔH_d of cross-linked microspheres decreases with increasing concentration of cross-linker. Furthermore, the decomposition of chitosan was observed at 324 °C, and the endothermic peaks for microspheres cross-linked with 0.5%, 1%, and 2% genipin were found at 265 °C, 278 °C, and 289 °C, respectively. This systematic increase in denaturation temperature for microspheres with increasing concentration of cross-linker, reveal that the cross-linking enhances the thermal stability in microspheres. However, the peak of flurbiprofen disappeared in flurbiprofen-loaded microspheres, which suggested that the flurbiprofen entrapped in microspheres was molecularly dispersed.

XRD analysis was carried out to prove the results of DSC analysis. XRD patterns of flurbiprofen, chitosan and flurbiprofen-loaded microspheres cross-linked with 0.5%, 1%, and 2% genipin are shown in Fig. 5 (B). In the X-ray diffractograms of flurbiprofen, sharp

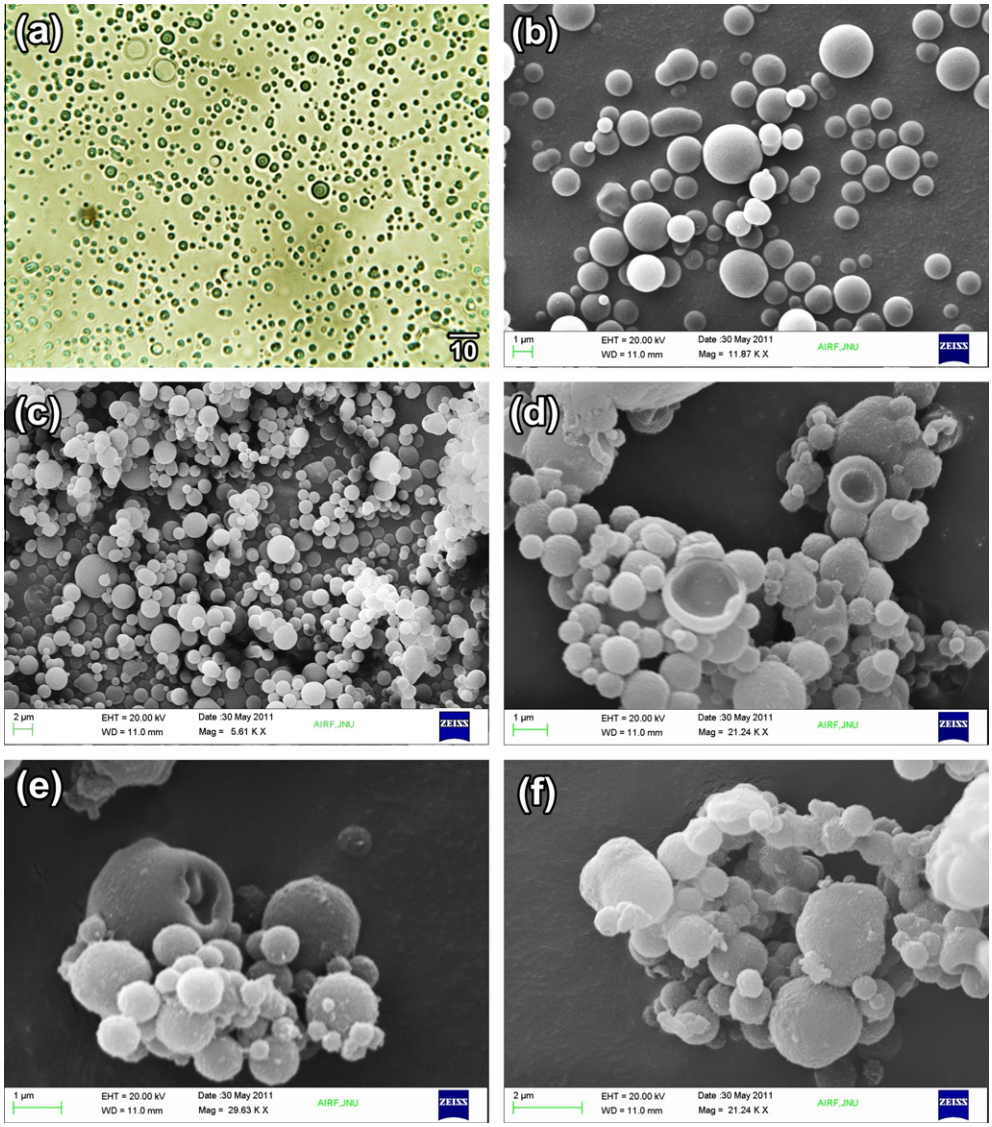


Fig. 2. Morphological features of chitosan microspheres: (a) light micrographic image; SEM images: (b) placebo uncross-linked microspheres; (c) CM-9 and after *in vitro* release study of 48 h (d) CM-7; (e) CM-8; and (f) CM-9. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

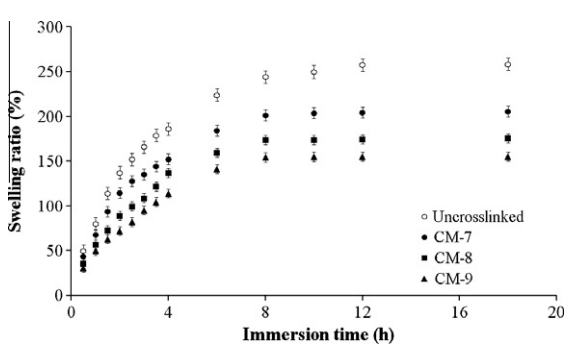
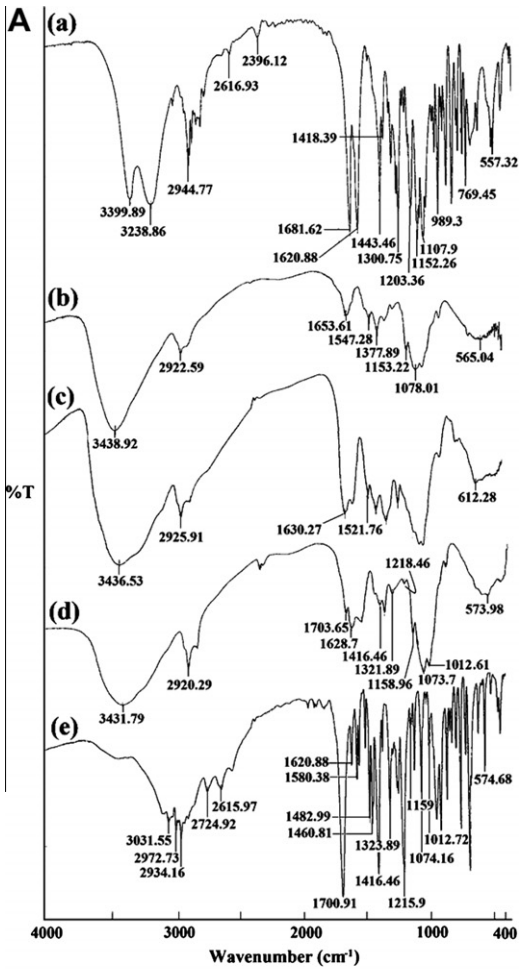


Fig. 3. Effect of different concentration of genipin on swelling ratio of chitosan microspheres. Values are mean ± SD ($n = 3$).

peaks at a different angle (2θ) of 7.19°, 10.27°, 19.53°, 21.47°, and 24.49° revealed the crystalline nature of drug, while flurbiprofen-loaded chitosan microspheres showed no sharp peaks. These results suggest that the flurbiprofen entrapped in the microspheres was in an amorphous state which is further supported by DSC studies. There was no significant difference was observed in the physical form of microspheres cross-linked with different concentration of genipin.

3.2. In vitro release studies

The results of *in vitro* drug release for flurbiprofen and formulations CM-7, CM-8, and CM-9 are shown in Fig. 6A. The complete *in vitro* dissolution of flurbiprofen (solubility in dissolution medium 4 ± 0.8 mg/mL) was attained within first 3 h. However, microspheres (CM-7, CM-8, and CM-9 cross-linked with 0.5%, 1%, and 2% genipin concentration, respectively) showed biphasic release pattern. The initial burst release was found to be 35–46% within the first 3 h. It might be attributed to flurbiprofen particles adsorbed on the surface of microspheres. In formulation CM-7, *in vitro* drug release was almost complete in 84 h, while CM-8 and CM-9 released about 86% and 78% of the total drug over the same period, respectively. It was observed that the drug release from CM-9 was slow and more controlled than CM-7 and CM-8. It may be noted that with an increase in cross-linking, the drug release



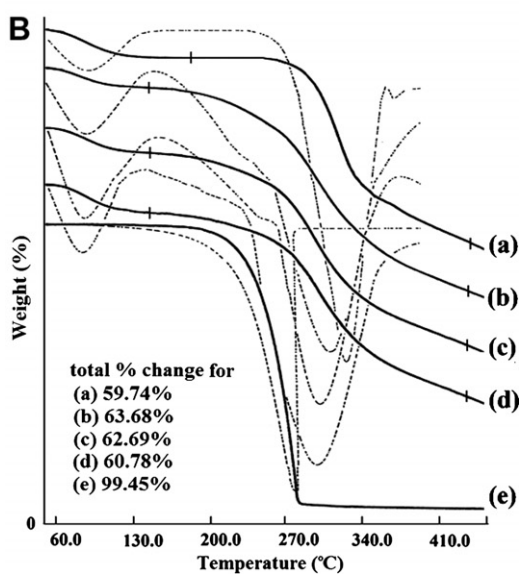


Fig. 4. FTIR spectra (A): (a) genipin, (b) chitosan, (c) placebo cross-linked microspheres, (d) flurbiprofen-loaded cross-linked microspheres, and (e) flurbiprofen. TGA thermograms (B): (a) chitosan, (b) flurbiprofen-loaded microspheres cross-linked with genipin at different concentration: (b) 0.5%, (c) 1%, and (d) 2% and (e) flurbiprofen.

decreases due to the rigidity of the polymer matrix that allows slow diffusion and erosion. These results are in accord with the results of swelling study. The SEM images (Fig. 2d–f) of chitosan

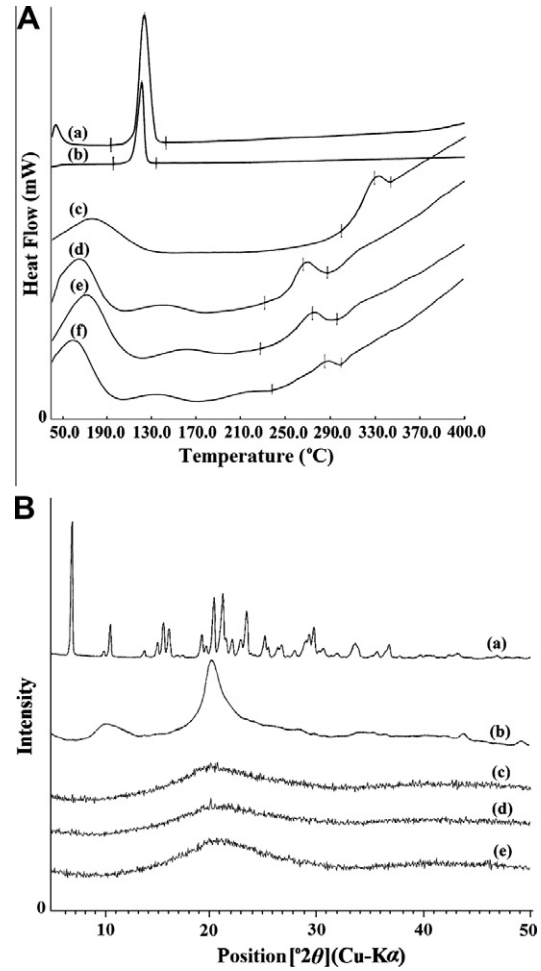


Fig. 5. DSC thermograms (A): (a) genipin, (b) flurbiprofen, (c) chitosan, flurbiprofen-loaded microspheres cross-linked with genipin at different concentration, (d) 0.5%, (e) 1%, and (f) 2%. X-ray diffractograms (B): (a) flurbiprofen, (b) chitosan, and (c) flurbiprofen-loaded microspheres cross-linked with genipin at different concentrations: (c) 0.5%, (d) 1% and (e) 2%.

microspheres after 48 h *in vitro* release study demonstrated that CM-9 cross-linked at higher concentration of genipin was hardly degraded, while noticeable surface degradation was observed for CM-7 and CM-8, cross-linked at lower concentration of genipin.

It is clear from Fig. 6B that the microspheres with more polymer concentration released the drug in a more controlled manner as compared to microspheres containing lesser polymer concentration. The release of flurbiprofen decreased with an increase in chitosan ratio, which may be due to increased density of polymer matrix and also an increase in the diffusional path length that the drug molecules have transverse [35]. The release profile of formulations prepared with 1:1, 1:2, and 1:3 drug-to-polymer ratio followed the order: CM-3 > CM-6 > CM-9, respectively.

Formulation CM-9 showed highest % drug entrapment (80.97 ± 2.18), better particle size distribution (6.14 ± 0.87) with smallest span value (1.03), regular and spherical shape as well as better controlled release pattern. Hence, it was selected as an optimized formulation for *in vivo* studies.

3.3. In vivo studies

Experimental animal models of arthritis are important for the understanding of pathology of human rheumatoid arthritis [36]. Twenty-four days after the induction of arthritis by seven i.a. injec-

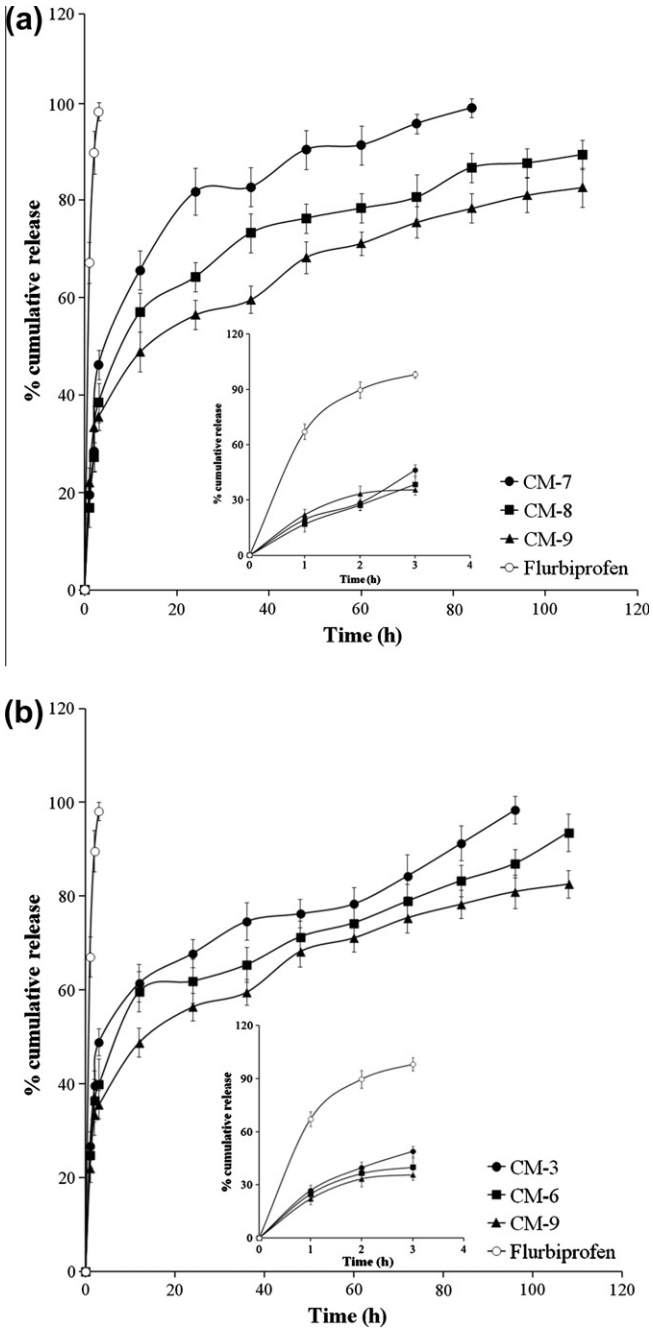


Fig. 6. *In vitro* drug release profile (A) of flurbiprofen and microspheres cross-linked with different genipin concentration (0.5%, 1%, and 2%) with burst release profile (inset) and (B) of flurbiprofen and microspheres prepared with different drug-to-polymer ratio (1:1, 1:2 and 1:3) with burst release profile (inset) (B). Values are mean ± SD ($n = 3$).

tions of carrageenan in the left knee joint of the rats, the difference in the right and left knee diameter was around 9.35 ± 0.67 mm. The histopathology images (Fig. 7) confirmed the carrageenan-induced arthritis in the rats.

The *in vivo* biocompatibility of the genipin cross-linked chitosan microspheres in the knee joint was evaluated. As shown in Fig. 7, histological examination of the knee joint after injection of saline in the right knee and microspheres suspension in saline in the left knee showed no inflammatory infiltrates. These results reported the biocompatibility of the i.a. injection of genipin cross-linked chitosan microspheres in the synovium.

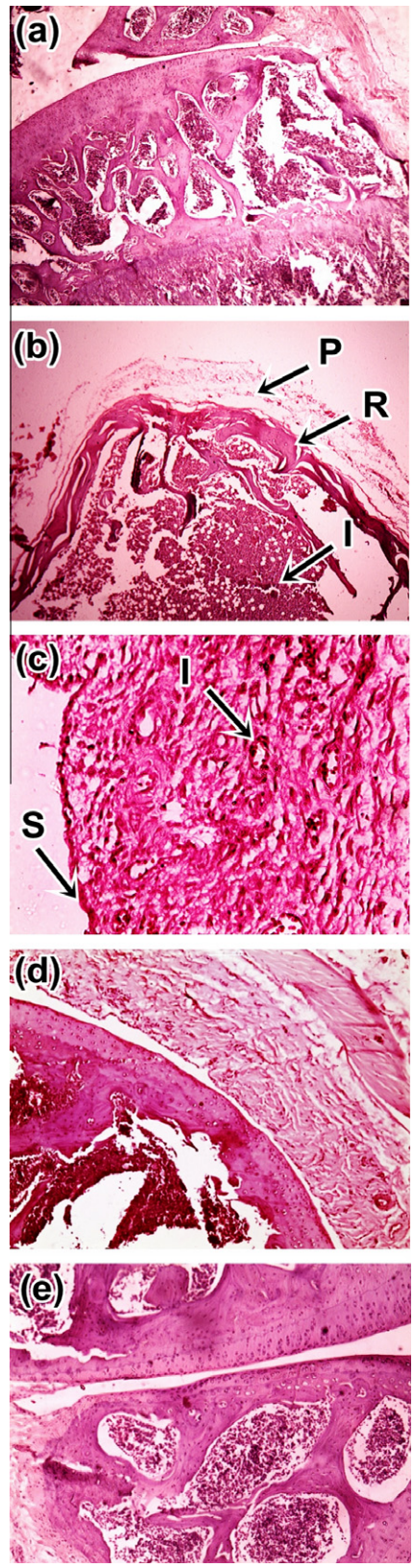


Fig. 7. Photomicrographs of histological slides of rat knee joints: (a) healthy joint, (b) arthritic joint, (c) arthritic joint (40× original magnification), (d) left knee joint injected with saline as control, and (e) right knee joint injected with genipin cross-linked chitosan microspheres. P = pannus overlying cartilage, R = ramnent of unmineralized cartilage, I = inflammatory infiltrate, S = synovial lining cell layer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

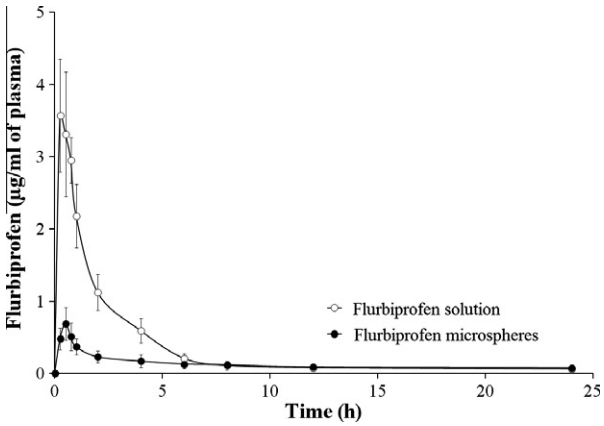


Fig. 8. Plasma concentration of flurbiprofen in arthritic rats after i.a. injection of flurbiprofen solution and its microspheres. Values are mean ± SD ($n = 6$).

Table 3
Mean (±SD) pharmacokinetic parameters of flurbiprofen in arthritic rats ($n = 6$) intra-articularly injected with flurbiprofen solution and its microspheres.

Pharmacokinetic parameters	Rats intra-articularly injected with	
	Flurbiprofen solution	Flurbiprofen microspheres
K_a (1/h)[a]	4.16 ± 1.276	1.21 ± 0.52[b]
C_{max} (µg/mL)	3.57 ± 1.34	0.69 ± 0.19[b]
T_{max} (h)	0.18 ± 0.09	0.39 ± 0.14[c]
$AUC_{(0-24)}$ (µg/mL/h)	8.62 ± 2.24	3.15 ± 1.12[b]
$AUC_{(0-\infty)}$ (µg/mL/h)	9.05 ± 2.68	4.02 ± 1.49[c]
MRT (h)	5.61 ± 1.36	13.5 ± 1.87[b]

[a] Absorption rate constant.
[b] Significant at $P < 0.05$.
[c] Significant at $P < 0.001$.

Table 4
Mean (±SD) of recovery of flurbiprofen (as % of administered dose) in arthritic knee joints (per 5 gm tissue) at different time intervals in rats ($n = 6$) intra-articularly injected with flurbiprofen solution and its microspheres.

Rats intra-articularly injected with	Time (h)		
	4	12	24
Flurbiprofen solution	0.0242 ± 0.0087	0.0161 ± 0.0094	0.0122 ± 0.0089
Flurbiprofen microspheres	0.0227 ± 0.0064	0.0596 ± 0.0165[a]	0.1071 ± 0.0697[a]

[a] Significant at $P < 0.05$.

3.3.1. Pharmacokinetic studies

The chromatograms of flurbiprofen confirmed a stable baseline and good resolution between flurbiprofen and plasma. The assay was linear in the range of 0.05–5 µg/ml. The mean absolute recovery in rat plasma was 91.21 ± 2.82%, and the mean relative recovery was 102.8 ± 3.21%. Intra- and inter-day accuracies were within an accuracy range of 95–110%. Intra- and inter-day precisions (<3%) were within the acceptance range. The mean residence time of flurbiprofen was 3.6 h.

The mean plasma concentration–time profiles of flurbiprofen following i.a. administration of the flurbiprofen solution and its microspheres are shown in Fig. 8, and the calculated pharmacokinetic parameters are presented in Table 3. It is evident from the figure that C_{max} (3.57 ± 1.34 µg/mL) was significantly higher and T_{max} (0.18 ± 0.09 h) was significantly smaller for flurbiprofen solution compared with its microspheres ($P < 0.05$ and $P < 0.001$, respectively). These results indicate rapid clearance of free drug from the knee joint of rat. The K_a (1.21 ± 0.52 h^{-1}) and $AUC_{(0-24)}$ (3.15 ± 1.12 µg/mL/h) were significantly lower, and MRT (13.5 ± 1.87 h) was significantly higher for microspheres compared with solution that showed the appreciable retention of drug in the knee joint for long time ($P < 0.05$). These pharmacokinetic data reveal that the genipin cross-linked chitosan microspheres are viable device to impede the absorption of effective drug concentration in blood as compared to the traditional drug solution that comes readily into the systemic circulation from arthritic knee joint. It confirms that i.a. administration of microspheres is competent to reduce the systemic effects of the treatment.

3.3.2. Evaluation of recovery of flurbiprofen in the rat knee joint

The arthritic rat knee joints injected with flurbiprofen solution and its microspheres were removed for the evaluation of recovery of flurbiprofen as the percent of administered dose. The recovery of flurbiprofen in arthritic knee joints (per 5 g tissue) at different time intervals are presented in Table 4. The recovery of flurbiprofen in rats at 4 h after i.a. injection of flurbiprofen solution (0.0242 ± 0.0083%) and its microspheres (0.0227 ± 0.0059%) was not significantly different. But the recovery values obtained for flurbiprofen microspheres were about 3.6 and 8.7 folds higher than its solution at 12 and 24 h, respectively. From the recovery data acquired from arthritic knee joint with flurbiprofen solution and its microspheres studied, it is evident that flurbiprofen is available at significantly high concentrations with microspheres and localization of the drug with microspheres was higher than the flurbiprofen solution. These results revealed the potential of microspheres to maintain high accumulation of drug for an extended time period in the arthritic rat knee joint.

4. Conclusions

Genipin cross-linked chitosan microspheres of flurbiprofen were successfully prepared using emulsification-cross-linking method. The drug entrapped in the microspheres was free from any chemical change. Genipin at higher concentration decreased the particle size and improved size distribution. It was observed that the drug release was dependent upon the cross-linking and drug-to-polymer ratio. The *in vitro* drug release from optimized microspheres was extended and in controlled manner. According to the results of *in vivo* studies, i.a. injection of these microspheres is safe to the synovium and maintains the drug concentration within arthritic rat knee joints significantly. It is concluded that genipin cross-linked chitosan microspheres have potential application as an i.a. drug delivery system.

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